

95.

$$\begin{array}{r} 6 \overline{) 0.40 \ 00 \ 00} \ (.63 \\ \underline{36} \\ 123 \underline{4 \ 00} \\ 3 \ 69 \end{array}$$

$$\therefore \sqrt{0.4} = 0.63$$

96.

$$\begin{array}{r} 4 \overline{) 0.20 \ 00 \ 00} \ (.447 \\ \underline{16} \\ 84 \underline{4 \ 00} \\ 3 \ 36 \\ 887 \underline{64 \ 00} \\ 62 \ 09 \end{array}$$

$$\therefore \sqrt{0.2} = 0.447$$

97.

$$\begin{array}{r} 3 \overline{) 0.12 \ 10 \ 00} \ (.347 \\ \underline{9} \\ 64 \underline{3 \ 10} \\ 2 \ 56 \\ 687 \underline{54 \ 00} \\ 48 \ 00 \end{array}$$

$$\therefore \sqrt{0.121} = 0.347$$

98.

$$\begin{array}{r} 2 \overline{) 0.06 \ 40 \ 00} \ (.252 \\ \underline{4} \\ 45 \underline{2 \ 40} \\ 2 \ 25 \\ 502 \underline{1500} \\ 1004 \end{array}$$

$$\therefore \sqrt{0.064} = 0.252$$

$$99. \sqrt{\frac{0.16}{0.4}} = \sqrt{\frac{0.16}{0.40}} = \sqrt{\frac{16}{40}} = \sqrt{\frac{4}{10}} = \sqrt{0.4} = 0.63.$$

$$100. \frac{1 + \sqrt{0.01}}{1 - \sqrt{0.1}} = \frac{1 + 0.1}{1 - 0.316} = \frac{1.1}{0.684} = \frac{1100}{684} = 1.6.$$

$$\begin{array}{r} 3 \overline{) 0.10 \ 00 \ 00} \ (.0316 \\ \underline{9} \\ 61 \underline{1 \ 00} \\ 61 \\ 626 \underline{39 \ 00} \\ 37 \ 56 \end{array}$$

$$101. \sqrt{535.9225} = \sqrt{\frac{5359225}{10000}} = \frac{2315}{100} = 23.15.$$

$$\begin{array}{r} 2 \overline{) 5 \ 35 \ 92 \ 25} \ (2315 \\ \underline{4} \\ 43 \underline{1 \ 35} \\ 1 \ 29 \\ 461 \underline{6 \ 92} \\ 4 \ 61 \\ 4625 \underline{231 \ 25} \\ 231 \ 25 \\ \times \end{array}$$

$$102. \frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5} = \frac{2.236}{5} = 0.447.$$

$$103. \sqrt{\frac{8}{3}} = \sqrt{\frac{8 \times 3}{3 \times 3}} = \frac{\sqrt{24}}{3} = \frac{4.899}{3} = 1.633.$$

$$104. \frac{3\sqrt{2}}{2\sqrt{3}} = \frac{3\sqrt{2}}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{6}}{2 \times 3} = \frac{\sqrt{6}}{2} = \frac{2.449}{2} = 1.2245.$$

$$\begin{aligned} 105. \frac{\sqrt{5}}{2} - \frac{10}{\sqrt{5}} + \sqrt{125} &= \frac{(\sqrt{5})^2 - 20 + 2\sqrt{5} \times 5\sqrt{5}}{2\sqrt{5}} \\ &= \frac{5 - 20 + 50}{2\sqrt{5}} = \frac{35}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{35\sqrt{5}}{10} = \frac{7}{2} \times 2.236 \\ &= 7 \times 1.118 = 7.826. \end{aligned}$$

$$106. \text{Clearly, } a * b = \sqrt{a^2 + b^2}. \therefore 5 * 12 = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

107. Let the missing digit be x .

$$\begin{array}{r} 1 \overline{) 1 \ 53 \ 7x} \ (124 \\ \underline{1} \\ 22 \underline{53} \\ 44 \\ 244 \underline{9 \ 7x} \\ 9 \ 76 \\ \times \end{array}$$

Then, $x = 6$.

$$\begin{aligned} 108. 7! &= 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 7 \times 2 \times 3 \times 5 \times 2^2 \times 3 \times 2 \\ &= 2^4 \times 3^2 \times 5 \times 7. \end{aligned}$$

Thus, the smallest perfect square number which is divisible by $7!$ is

$$(2^4 \times 3^2 \times 5 \times 7) \times (5 \times 7) = 5040 \times 35 = 176400.$$

$$109. \text{L.C.M. of } 3, 4, 5, 6, 8 \text{ is } 120. \text{ Now, } 120 = 2 \times 2 \times 2 \times 3 \times 5.$$

To make it a perfect square, it must be multiplied by $2 \times 3 \times 5$.

$$\text{So, required number} = 2^2 \times 2^2 \times 3^2 \times 5^2 = 3600.$$

$$\begin{aligned} 110. \text{L.C.M. of } 21, 36, 66 &= 2772. \text{ Now, } 2772 \\ &= 2 \times 2 \times 3 \times 3 \times 7 \times 11. \end{aligned}$$

To make it a perfect square, it must be multiplied by 7×11 .

$$\text{So, required number} = 2^2 \times 3^2 \times 7^2 \times 11^2 = 213444.$$

$$111. 294 = 7 \times 7 \times 2 \times 3.$$

To make it a perfect square, it must be multiplied by 2×3 i.e., 6.

$$\therefore \text{Required number} = 6.$$

$$112. 5808 = 2 \times 2 \times 2 \times 2 \times 3 \times 11 \times 11 = 2^2 \times 2^2 \times 3 \times 11^2.$$

To make it a perfect square, it must be multiplied by 3.

$$113. 1470 = 7 \times 7 \times 5 \times 6. \text{ To make it a perfect square, it must be divided by } 5 \times 6, \text{ i.e., } 30.$$